

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration: 1 Hour
Total Marks: 50

Annexure A

Scenario Based Assignment

Code of Conduct:

I Navin Kumar (C192511015) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Navin Kumar

19 + 19 + 22 + 18 + 12

796

HP

1. Smart Irrigation - Optimization problem

Formulation

The objective is to find the best settings for thousands of irrigation valves

- Variables: Valve opening levels.
- Inputs: Soil moisture weather and crop condition
- Objective: Minimize water wastage, energy use and crop stress.
- Constraints: Water availability, valve capacity, and crop requirement.

i) Hill climbing

- Hill climbing starts in local maxima/minima
- plateaus provide no clear improvement
- Ridges make the best direction difficult to find.
- Depends on the initial solution.

2. Autonomous Submarine - Online

i) Need for Online Search

The submarine does not have a complete map. It must explore, collect sonar information and make decisions at the same time.

ii) Challenges:-

- partial observability: sonar has limited range, so the complete environment is unknown.
- Dynamic environment: water currents and underwater conditions can change.
- Unknown obstacles can appear during exploration.

iii) Internal Models.

- detect obstacles
- Map explored areas.
- Records safe and unsafe locations.
- updates the map whenever new information

iv) Comparison

Search	Feature:-
Uniform cost	Uses path cost but can be slow
BFS	Finds shortest path but needs high memory
DFS	May explore unmitable paths.
A*	Efficient with a good heuristic
Online search	Works while exploring unknown areas.

v) Best Approach:-

An online search agent with continuous replanning is best because it can adapt new information, avoid hazards and make safe decisions in real time.

Hospital Nurse Scheduling - CSP

The nurse scheduling problem can be represented as a Constraint Satisfaction Problem (CSP)

1) Variable, Domains and Constraints.

Variable: Nurse assignments for each shift

Domain: Nurses who are available and qualified for the shift.

Constraints:

- No overlapping shifts
- Nurses cannot be assigned during leave.
- Only qualified nurse can work in ICU
- Senior nurses

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

81 + 20 + 25 + 20 + 15 = 101